

MAL3040 — Statistics
Assignment: Practice with Data

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About this sheet. There are fourteen questions. Part A is about making charts. Part B is about computing numbers from data. Part C is about what the data and the charts tell you. You do not need the textbook.

Please note. All the numbers on this sheet were made up for practice. They are not real measurements. So do not use any answer here as a real fact about students, workers, cricket or shops.

Formulas you will need. You have n observations $x_1, x_2, \dots, x_n$.

- Sample mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.
- Sample variance: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$. Always divide by $n-1$, never by n .
- Short cut for the variance: $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$.
- Sample standard deviation: $s = \sqrt{s^2}$.
- *Median*. Arrange the data from smallest to largest. If n is odd, the median is the middle value. If n is even, take the two middle values and average them.
- *Mode*. The value that occurs most often.
- *Range*. Largest value minus smallest value.

Worked example. Take the four numbers 2, 5, 5, 8. Here $n = 4$.

$$\sum x_i = 2 + 5 + 5 + 8 = 20, \text{ so } \bar{x} = 20/4 = 5.$$

$$\sum x_i^2 = 4 + 25 + 25 + 64 = 118.$$

$$\sum (x_i - \bar{x})^2 = 118 - 4 \times 5^2 = 118 - 100 = 18.$$

$$s^2 = 18/(4-1) = 6, \text{ and } s = \sqrt{6} = 2.45.$$

Sorted, the data is 2, 5, 5, 8. The two middle values are 5 and 5, so the median is 5. The mode is 5. The range is $8 - 2 = 6$.

Do your work in this style. Show $\sum x_i$ and $\sum x_i^2$ every time. Give answers correct to two decimal places.

Part A. Making charts

1. Twenty students were asked which subject they like most. The answers were: Mathematics 8 students, Physics 5 students, Chemistry 4 students, Biology 3 students.

- (a) Make a table with three columns: the subject, the frequency, and the relative frequency.
- (b) Draw a bar chart of the frequencies. Leave equal gaps between the bars.
- (c) For a pie chart, the slice of a subject with relative frequency r gets an angle of $360 \times r$ degrees. Add a fourth column to your table giving the angle for each subject. Add the four angles and check that you get 360° . Now draw the pie chart.

2. Twenty students wrote a test marked out of 10. Here are their marks.

6 7 5 8 6 9 7 6 4 8 7 5 6 8 7 9 6 5 7 6

- (a) Make a frequency table. Use four columns: the mark, the frequency, the relative frequency, and the cumulative relative frequency.
- (b) Draw a bar chart of the frequencies.

- (c) Draw the cumulative relative frequency graph. Mark the marks 4, 5, 6, 7, 8, 9 on the horizontal axis, and the numbers from 0 to 1 on the vertical axis.

3. The weights, in kilograms, of fifteen students are:

55 48 61 53 57 47 55 62 52 58 54 55 59 53 56

- (a) Draw a stem and leaf plot. Use the tens digit as the stem and the units digit as the leaf. So there will be three stems: 4, 5 and 6.
 (b) Count the leaves. You must get fifteen. Write down the count.
 (c) Now group the data into intervals of width 5, starting at 45. That is: 45–50, 50–55, 55–60, 60–65. Put a value into the interval that contains it, and put a value that falls exactly on a boundary into the interval on the right. Draw the histogram.

Part B. Computing from data

4. Take the five numbers 4, 7, 7, 9, 3.

- (a) Find $\sum x_i$ and $\sum x_i^2$.
 (b) Find the sample mean.
 (c) Find the median, the mode and the range.
 (d) Find the sample variance and the sample standard deviation. Use the short cut formula.

5. Use your frequency table from Question 2. Do not add the twenty marks one by one. Instead use the frequencies.

- (a) Add a column $f \times x$ to your table and find $\sum fx$. Then the sample mean is $\sum fx$ divided by 20. Compute it.
 (b) Add a column $f \times x^2$ and find $\sum fx^2$. Then use $\sum fx^2 - 20 \bar{x}^2$ to find the sample variance and the standard deviation.
 (c) Find the median and the mode of the twenty marks.

6. The daily wages of forty workers are grouped as follows.

Daily wage (rupees)	200–250	250–300	300–350	350–400	400–450
Number of workers	6	10	14	7	3

Here the exact wage of each worker is not known. Only the interval is known. So we assume that every worker in an interval earns the wage at the middle of that interval.

- (a) Write down the middle value of each of the five intervals.
 (b) Make a table with columns for the interval, the middle value x , the frequency f , and the product fx . Find $\sum fx$, and hence find the approximate mean daily wage.
 (c) Which interval has the largest frequency? This is called the modal class.
 (d) Add a cumulative frequency column. Which interval contains the twentieth worker, counting from the lowest wage? This is called the median class.

7. Twelve students were given a problem. The time taken by each student, in minutes, was:

8 12 15 9 20 11 14 10 18 13 16 12

- (a) Arrange the twelve times from smallest to largest.
 (b) Find the median. Then find the first quartile Q_1 and the third quartile Q_3 . Use this rule: to find the 100 p percentile, compute np ; if np is a whole number, average the np th and the $(np + 1)$ th smallest values; if it is not a whole number, round it up and take that value.
 (c) Find the interquartile range $Q_3 - Q_1$.
 (d) Write down the five-number summary: the smallest value, Q_1 , the median, Q_3 , and the largest value. Draw the box plot.

8. Take again the five numbers 4, 7, 7, 9, 3 of Question 4. You already know their mean, variance and standard deviation.

- (a) Add 10 to every number. Write down the new five numbers. Compute their mean, variance and standard deviation.
- (b) Now go back to the original five numbers and multiply each one by 2. Write down the new five numbers. Compute their mean, variance and standard deviation.
- (c) Compare with your answers to Question 4. Complete these two sentences. “If I add 10 to every observation, the mean ... and the standard deviation ...” “If I multiply every observation by 2, the mean ... and the standard deviation ...”
9. The monthly salaries of five workers in a small office, in thousands of rupees, are 18, 20, 22, 24, 26.
- (a) Find the mean salary and the median salary.
- (b) The manager also works in the office. His salary is 100 thousand rupees. Add him to the list, so there are now six salaries. Find the new mean and the new median.
- (c) How much did the mean change? How much did the median change?
- (d) A newspaper wants to report the “typical” salary in this office. Which of the two numbers should it use, and why?
10. The mean of six numbers is 15. Five of the numbers are 12, 18, 10, 20, 14. Find the sixth number. Show your reasoning in two steps: first find what all six numbers must add up to, then subtract.

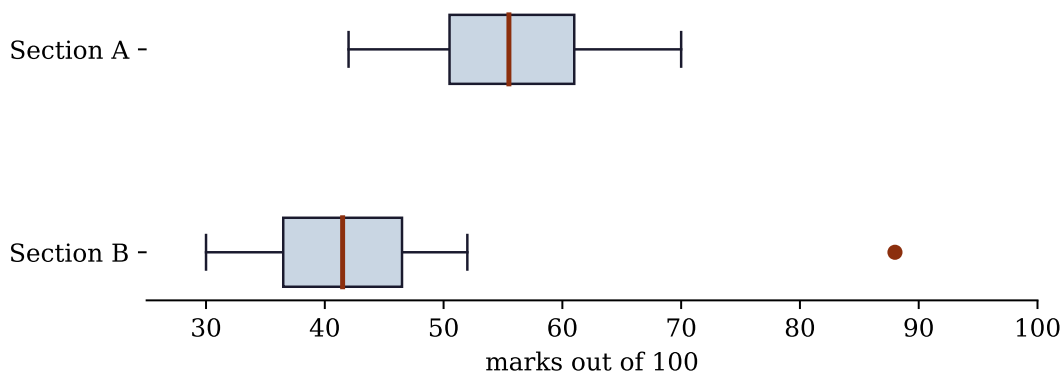
Part C. What the data tells you

Answer in full sentences. Questions 11 to 14 need very little calculation. Marks are for the explanation.

11. Two batsmen played five matches each. Their scores were:

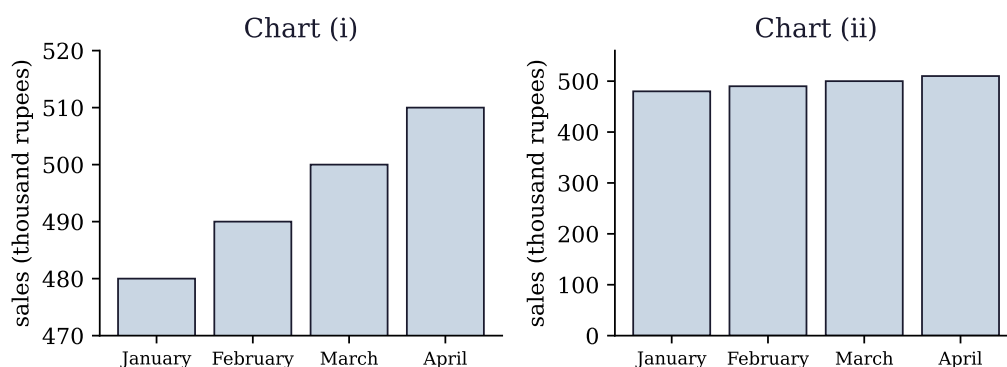
Batsman A	40, 42, 38, 41, 39
Batsman B	70, 10, 65, 5, 50

- (a) Find the mean score of each batsman. You will find that the two means are equal.
- (b) Find the standard deviation of each batsman’s scores.
- (c) The two batsmen have the same mean but very different standard deviations. Explain in one or two sentences what this difference tells you about how the two of them play.
- (d) Your team needs at least 35 runs from this batsman to win. Which one would you choose? Now suppose your team needs at least 60 runs. Which one would you choose now? Give a reason each time, using the actual scores.
12. The box plots below show the marks out of 100 of two sections of a class. Each section has twenty students.

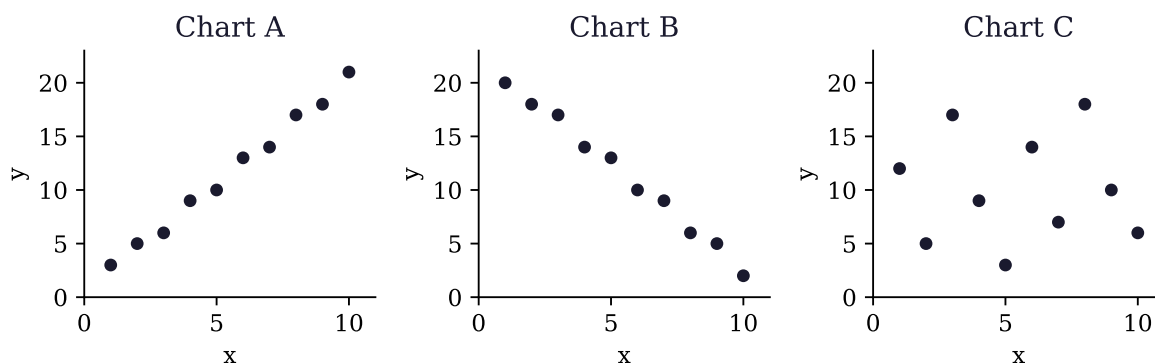


- (a) Which section has the higher median mark? Roughly what are the two medians?
- (b) In which section is the middle half of the students more spread out? How can you tell by looking?
- (c) In Section B, one mark is drawn as a separate dot away from the rest. What does that dot mean? Does it mean that a mistake was made while entering the marks?
- (d) A teacher says: “Section B has the best student in the class, so Section B is the stronger section.” Do you agree? Give a reason.

13. A shopkeeper drew two charts of his sales over four months. Both charts show exactly the same four numbers.



- There is only one difference between the two charts. What is it?
 - Look at Chart (i). By eye, roughly how many times taller is the April bar than the January bar? Now read the actual numbers off Chart (ii). By what factor did the sales really grow?
 - The shopkeeper wants to sell his shop and wants a buyer to believe that business is growing fast. Which chart will he show the buyer? Which chart is the honest one?
 - A bar shows its value by its *length*. Use this fact to explain in one sentence why the vertical axis of a bar chart should normally start at zero.
14. Below are three scatter diagrams. Each shows ten pairs of values.



- For each chart, say whether y tends to increase as x increases, or tends to decrease, or shows no clear pattern.
- In which chart do the points lie closest to a straight line? In which chart are they farthest from any straight line?
- In a certain town, somebody recorded two things in each of ten weeks: the number of ice creams sold, and the number of people who drowned while swimming. The scatter diagram looked like Chart A. The person concluded that eating ice cream causes drowning. Is this conclusion correct? If not, name a third quantity that could explain the pattern, and explain how it does so.

How to present your work. Draw lines for your tables. Label both axes of every chart and write the units. While computing, keep one extra decimal place, more than you plan to write in the final answer. A number written alone, with no sentence saying what it means, will not get full marks.